

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(A,B&C)

Time: 3HRS

Date: 29-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

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29-01-24_SR.STAR CO-SUPER CHAINA(A,B&C)_JEE-MAIN_GTM-30(SPCL)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYSICS			
CHEMISTRY			

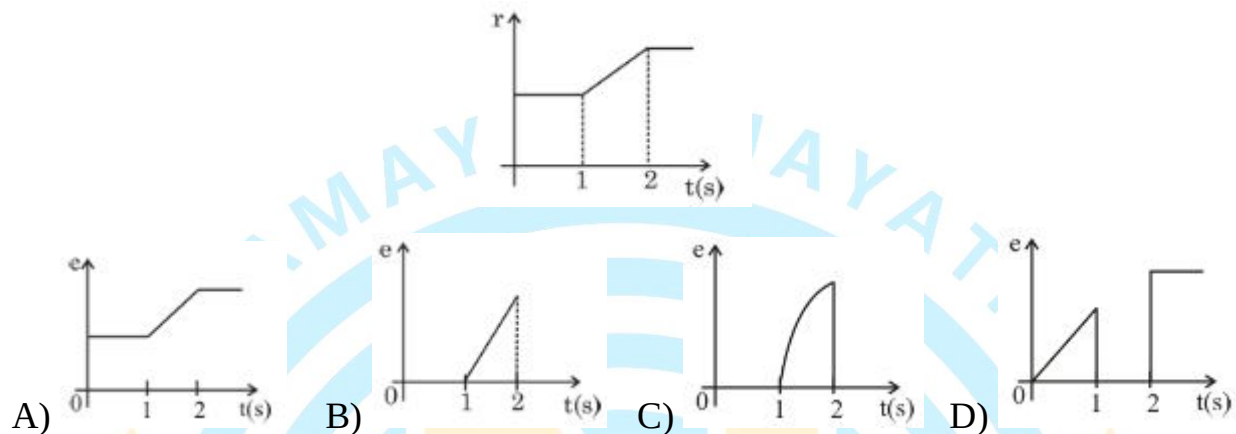
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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A flexible wire bent in form of circle is placed in uniform magnetic field perpendicular to plane of coil. Radius of coil changes as shown in figure. The graph of induced emf in coil is represented by :



2. Two resistances of equal magnitude R & having temperature coefficient α_1 and α_2 respectively are connected in parallel. The temperature coefficient of parallel combination is approximately.
- A) $2(\alpha_1 + \alpha_2)$ B) $\frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$ C) $\frac{\alpha_1 - \alpha_2}{2}$ D) $\frac{\alpha_1 + \alpha_2}{2}$
3. **Assertion :** When light is passed through a sample of hydrogen atoms in ground state, then wavelengths of absorption lines are same as wavelengths of lines of Lyman series in emission spectrum. and

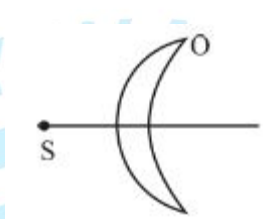
Reason : In ground state hydrogen atom will absorb only those radiation which will excite to higher energy level.

- A) Assertion is true, Reason is true and Reason is correct explanation for Assertion.
B) Assertion is true, Reason is true and Reason is NOT the correct explanation for Assertion.
C) Assertion is true, Reason is false.
D) Assertion is false, Reason is true.

4. A continuous flow water heater (geyser) has an electrical power rating = 2 kW & efficiency of conversion of electric power into heat = 80%. If water is flowing through the device at the rate of 100 cc/sec and the inlet temperature is 10°C , the outlet temperature is: ($S_w = 4200 \text{ J/kg } ^{\circ}\text{C}$)

A) 12.2°C B) 13.8°C C) 20°C D) 16.5°C

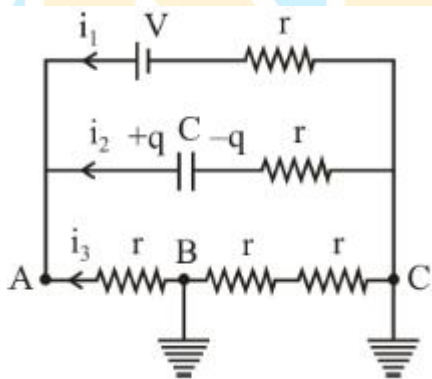
5. Optical instrument lens O and object S are placed along optic axis. Distance between object and lens can be varied. Properties of images can be :



(P) Real image (Q) Virtual image (R) Magnified Image (S) Image at ∞

A) P, Q, S B) Q, R C) P, Q, R D) P, Q, R, S

6. In the circuit below point B and C are grounded. Select the correct option.



A) $i_1 = \frac{V}{r}$ B) $i_2 = \frac{V}{2r}$ C) $i_3 = \frac{V}{2r}$ D) $q = \frac{CV}{2}$

7. An object of density d_0 kept deep inside water of density d_w . It is then released from rest. During the time it moves a vertical distance h within the water.

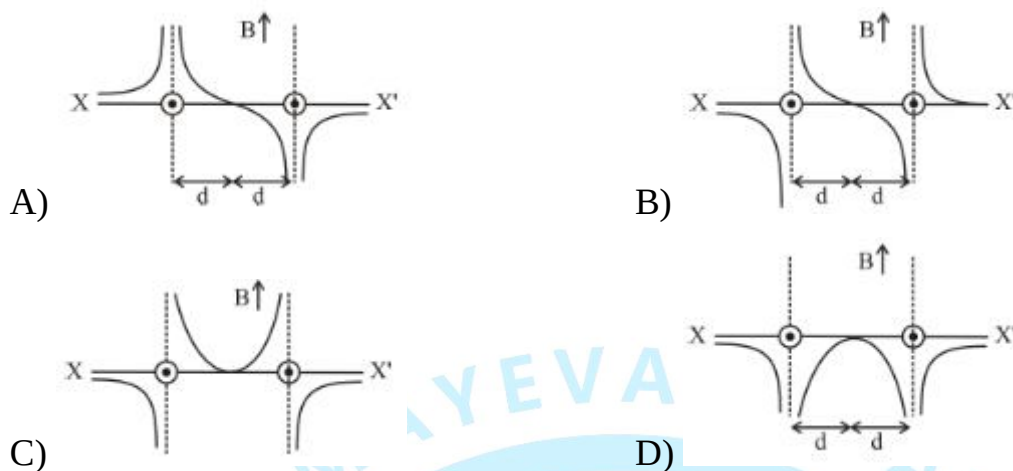
A) The gravitational potential energy of the water in vessel increase if $d_0 < d_w$.

B) The gravitational potential energy of the water in vessel decreases if $d_0 < d_w$.

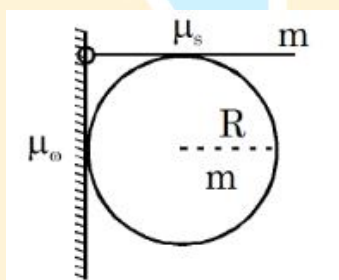
C) The gravitational potential energy of object increases if $d_0 > d_w$.

D) The gravitational potential energy of the object decreases if $d_0 < d_w$.

8. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper, as shown. The variation of the magnetic field B along the XX' is given by :



9. A stick with mass m and length $2R$ is pivoted at one end on a vertical wall. It is held horizontal, and a disc with mass m and radius R is placed beneath it, in contact with both, it and wall as shown in figure. The coefficient of friction between disc and wall is μ_w , between disc and stick is μ_s . System is released and it does not fall. Now match list-I and II.



	List-I		List-II
(A)	$(\mu_s)_{\min}$	(P)	1
(B)	$(\mu_w)_{\min}$	(Q)	2
(C)	Vertical component of contact force between stick and disc	(R)	2 mg
(D)	Vertical component of contact force between wall and disc	(S)	mg

A) (A) - (Q) ; (B) - (P) ; (C) - (S) ; (D) - (R)

B) (A) - (R) ; (B) - (P) ; (C) - (S) ; (D) - (Q)

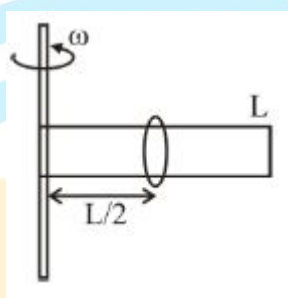
C) (A) - (P) ; (B) - (S) ; (C) - (Q) ; (D) - (R)

D) (A) - (S) ; (B) - (P) ; (C) - (R) ; (D) - (Q)

10. Find distance of a point from Earth's centre where the resultant gravitational field due to Earth & Moon is zero. (Given : mass of Earth : 6×10^{24} kg & mass of Moon : 7.4×10^{22} kg) The distance between Earth & Moon is 4.0×10^5 km.

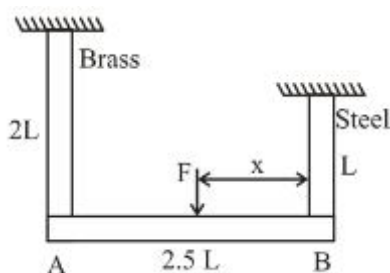
A) 3.6×10^5 km B) 7.2×10^5 km C) 3.6×10^4 km D) 7.2×10^4 km

11. At $t = 0$ a bead of mass m is placed at a distance $\frac{L}{2}$ from the axis of rotation. Initial speed of the bead is zero with respect to the rod. The rod is rotated with a constant angular velocity ω as shown in diagram. Find the velocity of bead w.r.t ground, when it leaves out of the rod. (Given L is the length of rod.)



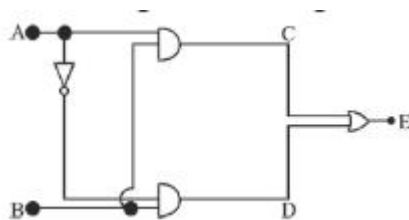
A) $\frac{\sqrt{3}\omega L}{2}$ B) $\frac{\sqrt{5}\omega L}{2}$ C) $\frac{\sqrt{7}\omega L}{2}$ D) $\frac{\sqrt{9}\omega L}{2}$

12. A very stiff bar (AB) of negligible mass is suspended horizontally by two vertical rods as shown in figure. Length of bar is $2.5L$. The steel rod has length L and cross-sectional radius of r and brass rod has length $2L$ & cross-sectional radius of $2r$. A vertically downward force F is applied to bar at a distance of x from steel rod and the bar remains horizontal. Find the value of x if ratio of young's modulus of steel and brass is $\frac{\gamma_s}{\gamma_B} = 2$.



(A) $1.5 L$ (B) L (C) $1.7 L$ (D) $1.25 L$

13. Truth table for the given circuit figure is :-



A	B	E	A	B	E	A	B	E	A	B	E
0	0	1	0	0	1	0	0	0	0	0	0
0	1	0	0	1	0	0	1	1	0	1	1
1	0	1	1	0	0	1	0	0	1	0	1
A) 1	1	0	B) 1	1	1	C) 1	1	1	D) 1	1	0

14. A series LR circuit is connected to an ac source of frequency ω and the inductive reactance is equal to $2R$. A capacitance of capacitive reactance equal to R is added in series with L and R . The ratio of the new power factor to the old one is :

- A) $\sqrt{\frac{2}{5}}$ B) $\sqrt{\frac{2}{3}}$ C) $\sqrt{\frac{3}{2}}$ D) $\sqrt{\frac{5}{2}}$

15. **Statement-1** : When light falls on a sphere made of diamond, total internal reflection takes place which makes it shine more than the similar sphere made of common glass.

Statement-2 : Refractive index for diamond is more than refractive index of cheap glass.

A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

B) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.

C) Statement-1 is True, Statement-2 is False.

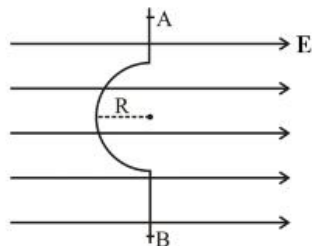
D) Statement-1 is False, Statement-2 is True.

16. Three point charges $+q, -2q$ and $+q$ are placed at points

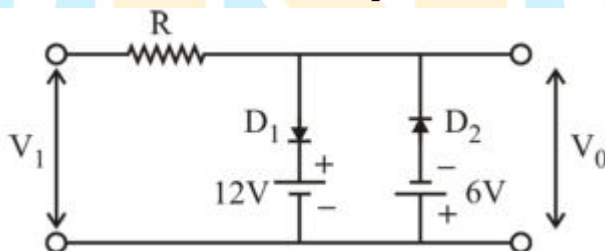
$(x=0, y=a, z=0), (x=0, y=0, z=0)$ and $(x=a, y=0, z=0)$ respectively. The resultant dipole moment can be best represented by $qa(\alpha\hat{i} + \beta\hat{j})$. Find $\alpha + \beta$.

- A) 2 B) 3 C) 4 D) 5

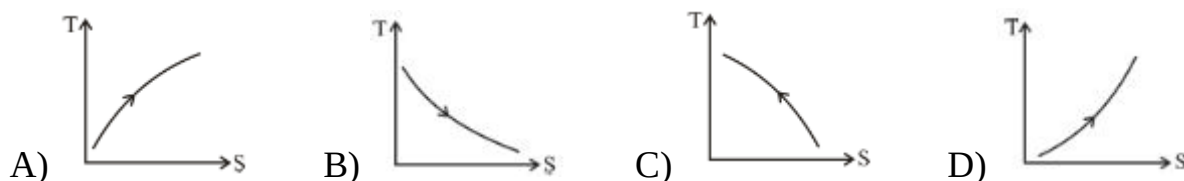
17. Figure shows a wire, hinged at A and B, and given the shape of half ring of radius R, the assembly is now free to rotate about an axis passing through AB in a uniform electric field E as shown in the figure. Linear charge density of wire is λ_1 and linear mass density is λ_2 . If assembly is slightly disturbed from unstable equilibrium position, then find maximum angular velocity of ring during ensuing motion.



- A) $\sqrt{\frac{16\lambda_1 E}{\pi\lambda_2 R}}$ B) $\sqrt{\frac{\pi \lambda_1 E}{16 \lambda_2 R}}$ C) $\sqrt{\frac{\lambda_1 E}{2\lambda_2 R}}$ D) $\sqrt{\frac{8\lambda_1 E}{\pi\lambda_2 R}}$
18. A thin oil film of refractive index 1.2 floats on the surface of water ($\mu = \frac{4}{3}$). When a light of wavelength $\lambda = 9.6 \times 10^{-7} \text{ m}$ falls normally on the film from air, then it appears dark when seen normally. The minimum change in its thickness for which it will appear bright in normally reflected light by the same light is :-
- A) 10^{-7} m B) $2 \times 10^{-7} \text{ m}$ C) $3 \times 10^{-7} \text{ m}$ D) $5 \times 10^{-7} \text{ m}$
19. A sinusoidal peak voltage of 15V connected between input terminals of circuit shown in figure. Assume diodes are ideal. In the output wave form,



- A) Positive peaks of input clipped at +12V and negative peaks at -6V
 B) Positive peaks of input clipped at +6V and negative peaks at -12 V
 C) Positive peaks of input clipped at +12V and negative peaks at -12 V
 D) Positive peaks of input clipped at +6V and negative peaks at -6 V
20. An ideal gas undergoes isobaric expansion. Graph between temperature of gas and entropy is :-



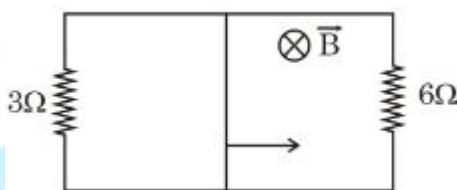
SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

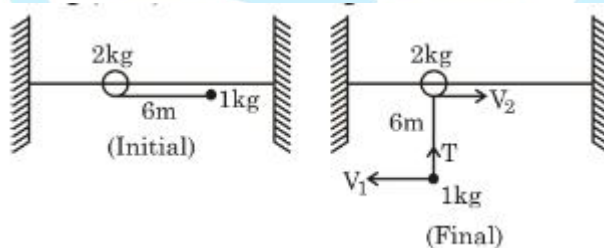
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

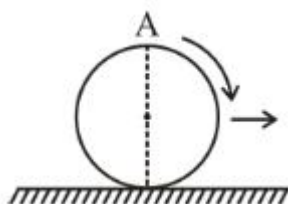
21. A rectangular loop and a sliding connector of length $\ell = 1\text{m}$ is located in uniform magnetic field $B = 2\text{T}$ perpendicular to plane of loop. Resistance of connector is $r = 2\Omega$. Two resistances of 6Ω and 3Ω are connected as shown in figure. Find external force (in N) required to hold the loop stationary as connector moves with constant velocity $v = 2\text{m/s}$.



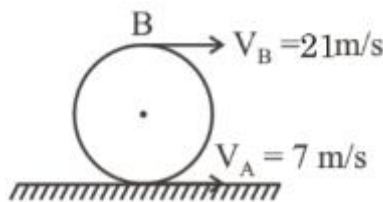
22. Kinetic energy of a body is increased by 21%. What is the percent increase in magnitude of linear momentum of the body?
23. An object is displaced from $\vec{r}_1 = (2\hat{i} + 3\hat{j})\text{m}$ to $\vec{r}_2 = (4\hat{i} + 6\hat{j})\text{m}$ under action of a force $\vec{F} = (3x^2\hat{i} + 2y\hat{j})\text{N}$. Find work done by the force (in J).
24. A bead of mass 2 kg is free to move along a smooth horizontal rod connected between two fixed points. The bead is connected to a mass of 1 kg by string of length 6 m as shown in the diagram. The mass is released from rest when the string is horizontal. Find the tension in the string (in N) when the string becomes vertical.



25. A ring is rolling, without slipping on a horizontal surface with constant velocity. Speed of point A at top is V_A . After an interval T, speed of point A again becomes V_A . The fraction of the interval T when speed of point A was greater than $\frac{\sqrt{3}V_A}{2}$ is T/x . Find x.



26. A solid sphere of radius R is released on a rough horizontal surface with its top point having thrice the velocity of its bottom point A ($V_A = 7 \text{ m/s}$) as shown in figure. Calculate linear velocity (in m/s) of centre of sphere when it starts pure rolling.



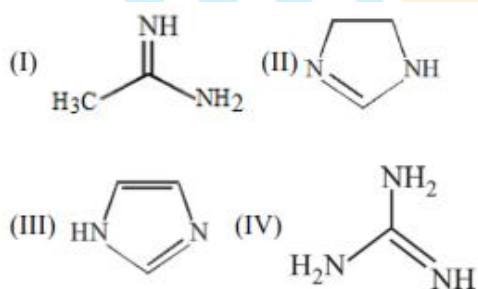
27. A point source of sound is located inside sea water. Bulk modulus of sea water is $B_w = 2.0 \times 10^9 \text{ N/m}^2$. A diver located at distance of 10 m from source registers a pressure amplitude of $\Delta P_0 = 3000\pi \text{ N/m}^2$ and equation of sound wave as $y = A \sin(1.5\pi x - 2100\pi t)$, y and x are in meter and t in second. Find the power (in kW) of the sound source.
(Take : $\pi = \frac{22}{7}$)
28. An ideal monoatomic gas A is supplied heat so as to expand without changing its temperature. In another process, starting with same state, it is supplied heat at constant pressure. In both cases a graph of work done by gas is plotted versus heat added (Q) to the gas. The ratio of slope of graphs obtained in first & second process is η_1 . The same ratio obtained for ideal diatomic gas is η_2 . If the ratio $\frac{\eta_1}{\eta_2} = \frac{a}{b}$. Then find $a + b$.
29. Nucleus ${}_3\text{A}^7$ has binding energy per nucleon of 10 MeV . It captures a proton and its mass is increased by $\frac{99}{100}$ times mass of proton. If binding energy of the new nucleus formed is $\alpha \text{ MeV}$ then find the value of 10α . [Mass energy of proton is 930 MeV]
30. A power cable of copper is stretched (initial tension zero) straight between two fixed towers. If the temperature decreases, the cable tends to contract. The amount of contraction for a free copper cable or rod is 0.02% per degree Celsius. Estimate what temperature decrease (in $^\circ\text{C}$) will cause the cable to snap. Assume that the cable obeys Hooke's law until it reaches its breaking point, which for copper occurs at a tensile stress of $2.2 \times 10^8 \text{ N/m}^2$. Ignore the weight of the cable and the sag and stress produced by the weight. Young's modulus for copper is $1.1 \times 10^{11} \text{ N/m}^2$.

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

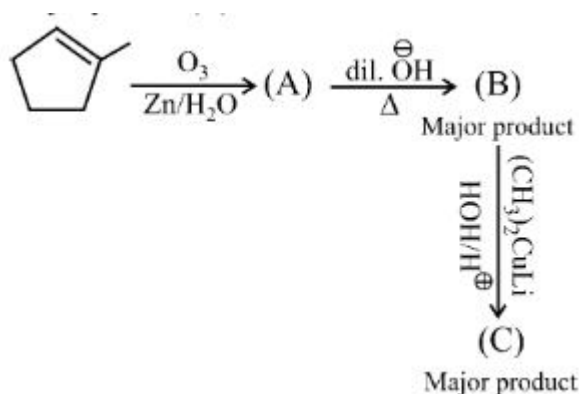
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

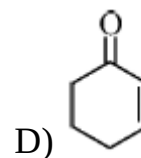
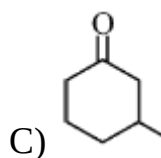
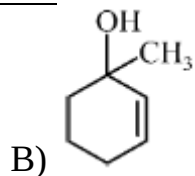
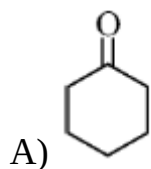
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. Which of the following is **INCORRECT** for f-block elements
- A) For lanthanoids +3 is the most common oxidation state.
 B) Ce^{+3} is more stable than Ce^{+4}
 C) Basic character of oxides of lanthanoids decreases on increasing atomic number.
 D) Lanthanoid elements form only LnC_2 type of carbide when heated with Carbon
32. Which of the following order is **INCORRECT** with respect to the given property?
- A) $\text{NaF} < \text{NaCl} < \text{NaBr} < \text{NaI}$: Ionic character
 B) $\text{Li}^+(\text{aq.}) > \text{Na}^+(\text{aq.}) > \text{K}^+(\text{aq.})$: Hydrated radius
 C) $\text{BeCO}_3 < \text{MgCO}_3 < \text{CaCO}_3$: Thermal stability
 D) $\text{AgI} > \text{AgBr}$: Covalent character
33. The order of basicity among the following compounds is

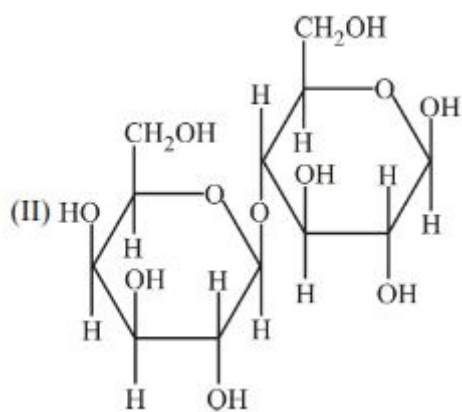
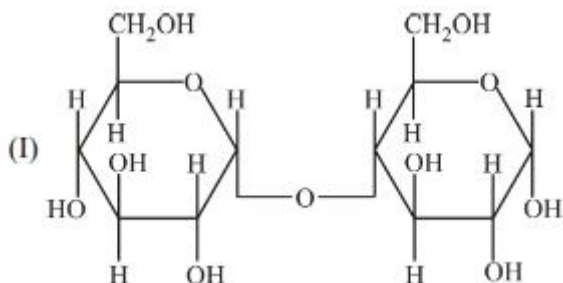


- A) $\text{II} > \text{I} > \text{IV} > \text{III}$ B) $\text{IV} > \text{II} > \text{III} > \text{I}$
 C) $\text{I} > \text{IV} > \text{III} > \text{II}$ D) $\text{IV} > \text{I} > \text{II} > \text{III}$
34. Major product (C) is :



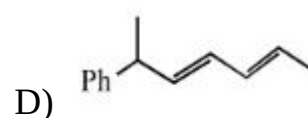
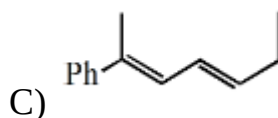
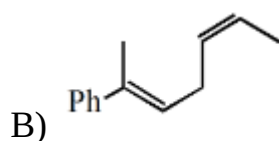
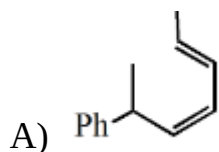
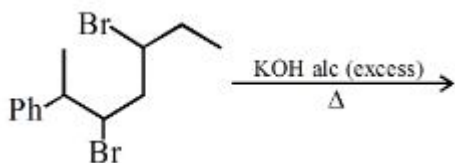


35. Correct statement about I and II

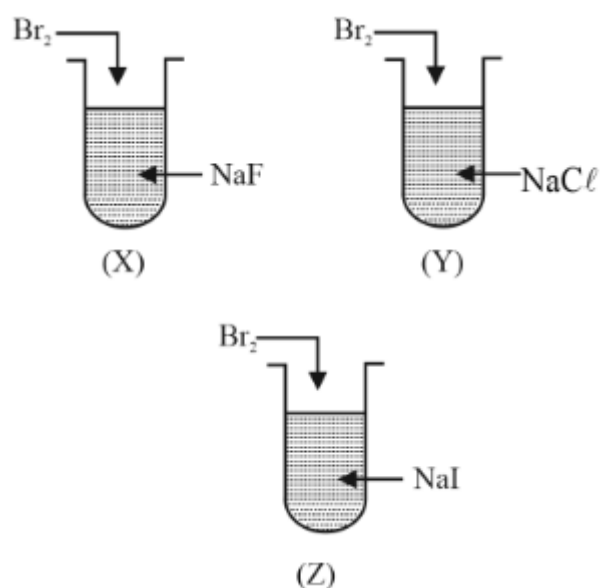


- A) I is reducing sugar
B) II is reducing sugar
C) I and II both are reducing sugars
D) None of the two are reducing sugars

36. The major product of the following reaction is

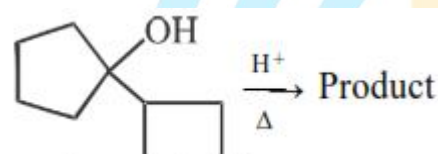


37. Which is the correct observation when is treated with NaF, NaCl and NaI taken in three test tubes labelled as (X), (Y) and (Z)?

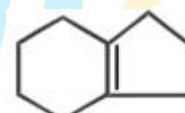
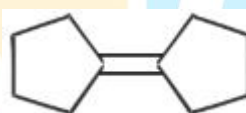
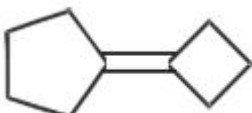
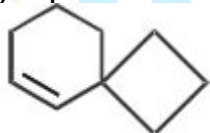


- A) F_2 is liberated in (X) and Cl_2 in (Y) B) Only I_2 is liberated in (Z)
 C) Only Cl_2 is liberated in (Y) D) Only F_2 is liberated in (X)

38.



Major product is



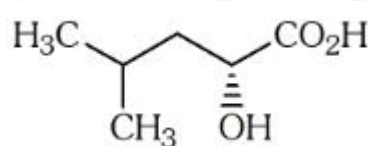
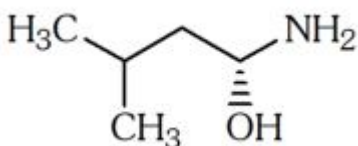
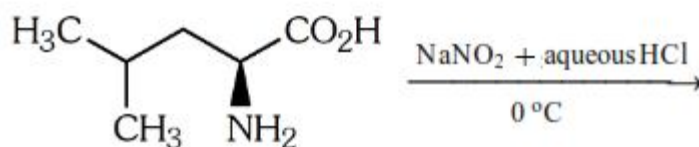
A)

B)

C)

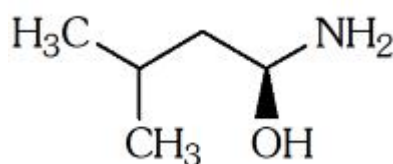
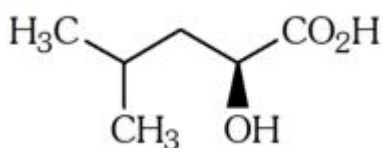
D)

39. The major product of the reaction is



A)

B)



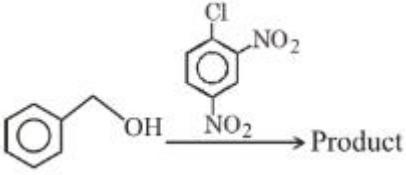
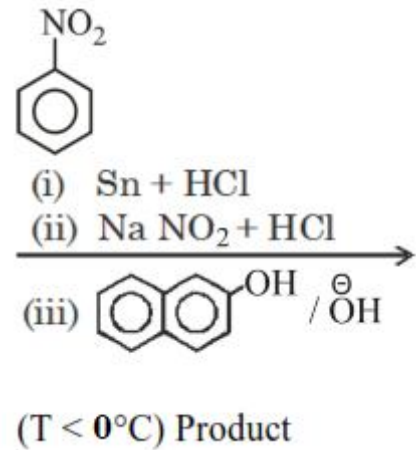
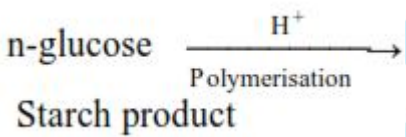
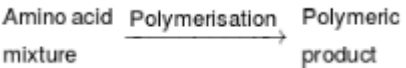
C)

D)

40. Assertion (A) : Helium atom has highest ionisation energy among all the elements.
Reason (R) : Helium has positive electron gain enthalpy.

- A) A is false but R is true.
B) A is true but R is false.
C) Both A and R are true and R is the correct explanation of A.
D) Both A and R are true and R is NOT the correct explanation of A.

41.

	List-I		List-II
(P)		(1)	Product give molish test
(Q)		(2)	Product give violet colour with neutral FeCl_3
(R)		(3)	L.S. of product give blue colour solution with FeSO_4 solution
(S)		(4)	Product give Biurate test

- (A) $P \rightarrow 3; Q \rightarrow 2,3; R \rightarrow 1; S \rightarrow 4$
 (B) $P \rightarrow 2; Q \rightarrow 2,3; R \rightarrow 1; S \rightarrow 3,4$
 (C) $P \rightarrow 3; Q \rightarrow 2,3; R \rightarrow 1; S \rightarrow 3,4$
 (D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1,3$

42. The radius of the second Bohr orbit of Li^{+2} ion, in terms of the Bohr radius (a_0) of hydrogen atom is :
- A) $\frac{4a_0}{9}$ B) $\frac{2a_0}{9}$ C) $\frac{2a_0}{3}$ D) $\frac{4a_0}{3}$
43. The correct order for energy of absorption in the following metal complexes is
A: $[\text{Ni}(\text{en})_3]^{2+}$, B: $[\text{Ni}(\text{NH}_3)_6]^{2+}$, C: $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
- A) $C < B < A$ B) $B < C < A$ C) $C < A < B$ D) $A < C < B$
44. Acetic acid exists in benzene solution in the dimeric form. In an actual experiment the Van't Hoff factor of acetic acid in benzene solution was found to be 0.52. Calculate the degree of association and osmotic pressure of 0.1M acetic acid solution in benzene at 300K
- A) 0.96 and 9.6 atm B) 0.96 and 1.28 atm
C) 0.46 and 4.83 atm D) 0.26 and 7.86 atm
45. $\text{A}(\text{g}) \rightarrow \text{B}(\text{g}) + 3\text{C}(\text{g})$
- In a closed container at a given temperature, pressure increases from 100 mm Hg to 160 mm Hg in 10 seconds, for the given reaction. Then the average rate of reaction in first 10 sec. will be :
- A) 2 mm/sec B) 4 mm/sec C) 6 mm/sec D) 3 mm/sec
46. The only **INCORRECT** option among the following is :
- A) In isothermal expansion of an ideal gas, internal energy of the gas remains unchanged.
- B) In adiabatic free expansion of an ideal gas, the internal energy of gas remains unchanged.
- C) In adiabatic free expansion of an ideal gas, the initial and final temperature of the gas remains same.
- D) In isobaric expansion of an ideal gas, the internal energy of gas remains unchanged.

47. The following reaction is performed at 298 K. $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, $K_p = 1.6 \times 10^{12}$

The standard free energy of formation of $\text{NO}(\text{g})$ is 86.6 kJ/mol at 298 K. What is the standard free energy of formation of $\text{NO}_2(\text{g})$ (in J mol^{-1}) at 298 K ?

A) $2 \times 86,600 - R(298) \ln(1.6 \times 10^{12})$

B) $0.5 \left[2 \times 86,600 - R(298) \ln(1.6 \times 10^{12}) \right]$

C) $R(298) \ln(1.6 \times 10^{12}) - 86600$

D) $86600 + R(298) \ln(1.6 \times 10^{12})$

48. Given below are two statements.

Statement I: In the titration between a strong acid and a weak base methyl orange may be suitable as an indicator.

Statement II: For titration of acetic acid with NaOH , phenolphthalein is not a suitable indicator. In the light of the above statements, choose the most appropriate answer from the options given below:

A) Statement I is false but Statement II is true

B) Statement I is true but Statement II is false

C) Both Statement I and Statement II are true

D) Both Statement I and Statement II are false

49. For combustion of ethanol $\text{C}_2\text{H}_5\text{OH}_{(\text{l})} + 3\text{O}_{2(\text{g})} \rightarrow 2\text{CO}_{2(\text{g})} + 3\text{H}_2\text{O}_{(\text{l})}$, the amount of heat produced as measured in bomb calorimeter, is 1364.47 kJ/mol at 25°C . Assuming all gases are behaving ideally, the enthalpy of combustion $\Delta_c H$, for ethanol will be :

($R = 8.314 \text{ kJ/mol}$)

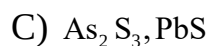
A) -1366.95 kJ/mol

B) -1361.95 kJ/mol

C) -1460.50 kJ/mol

D) -5.50 kJ/mol

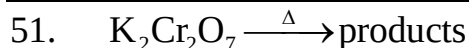
50. Which option contain both yellow colored sulphide precipitates?



SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

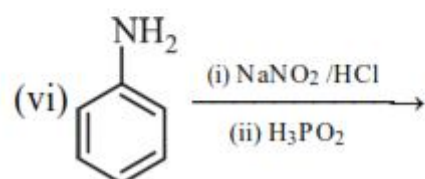
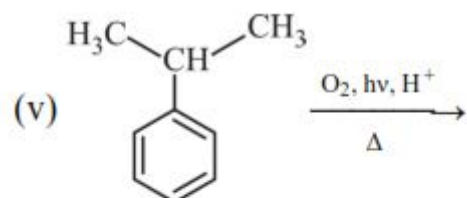
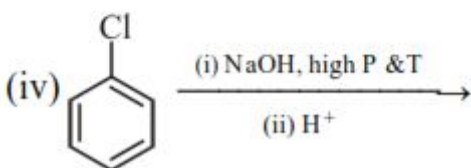
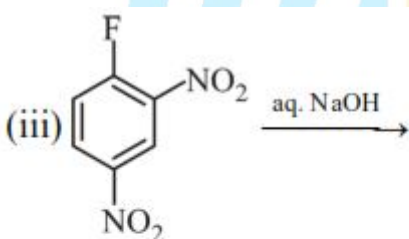
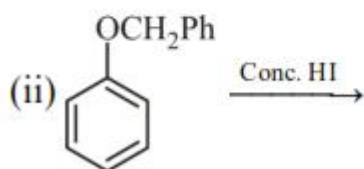
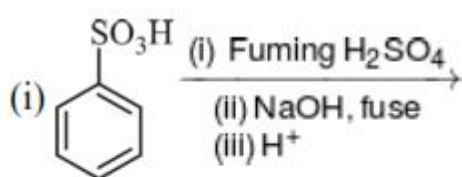
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

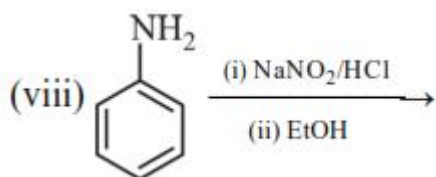
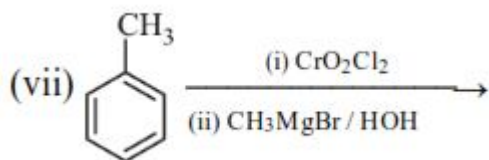
Marking scheme: +4 for correct answer, -1 in all other cases.



Let the sum of oxidation states of Cr in the different products be X. Find (X)

52. In how many reactions given below phenolic group is present in the major product:





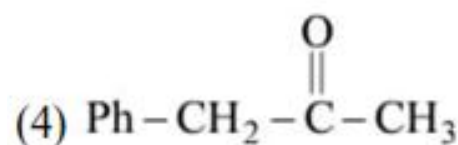
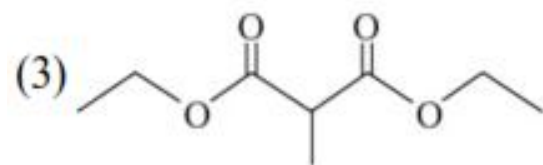
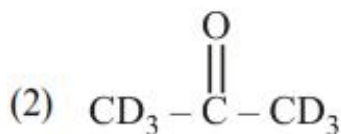
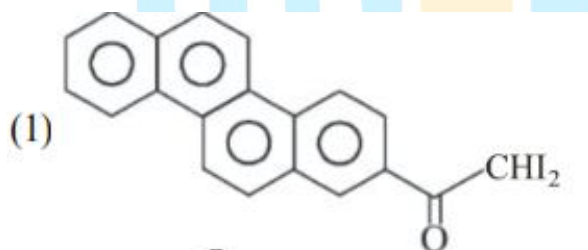
53. The reduction potential of M^{+3} and M couple at $pH = 12$ so that sparingly soluble salt $M(OH)_3$ will just start precipitation at 298 K is 'V' volt, then $100 \times V$ will be :

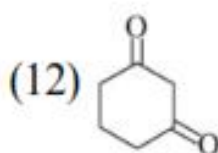
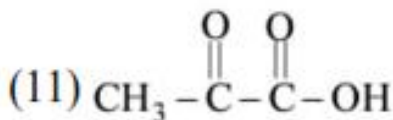
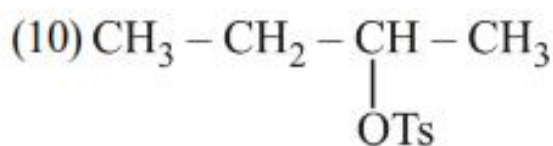
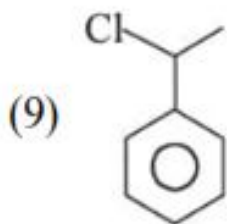
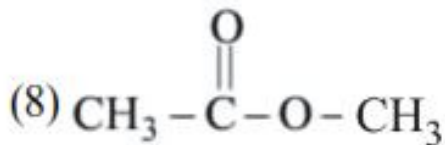
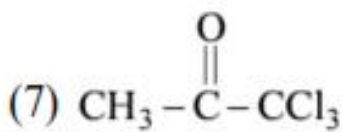
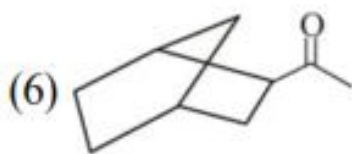
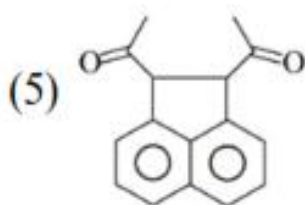
Given : $E_{M^{+3}/M}^0 = +0.40 \text{ V}$

K_{sp} of $M(OH)_3 = 1 \times 10^{-18} \text{ M}^4$

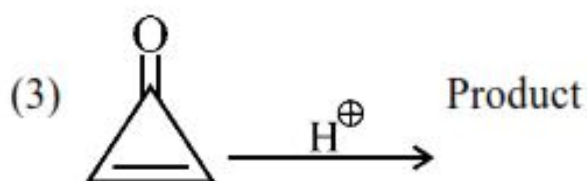
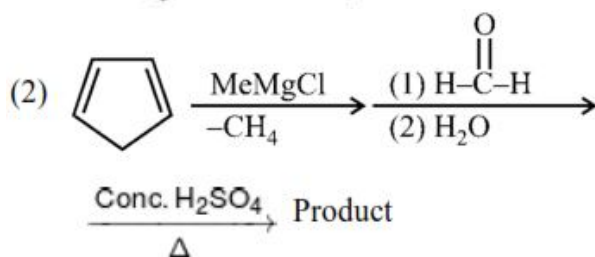
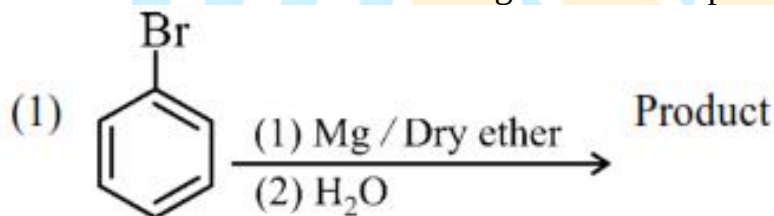
[Take $\frac{2.303RT}{F} = 0.06$]

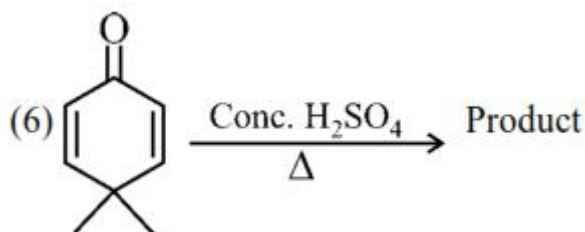
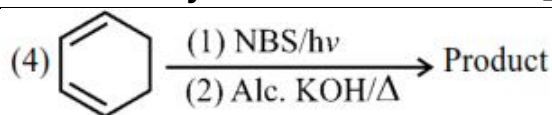
54. Which of the following compound (s) gives haloform reaction (or Iodoform test with $\text{NaOH} + \text{I}_2$)





55. Total number of reactions which give aromatic product as a major product ?





56. When the solution containing Ferric-chloride is treated with potassium ferrocyanide solution, a blue precipitate 'A' is obtained. Find the number of atoms in one molecule of 'A'.
57. Out of given molecules total molecules which are non polar.
 $\text{IOF}_5, \text{H}_2\text{O}_2, \text{ClF}_3, \text{CO}_2, \text{SO}_2, \text{CH}_2(\text{CN})_2, \text{C}_2(\text{CN})_4, (\text{CN})_2$
58. A mixture containing 2 moles of $\text{K}_3\text{H}(\text{C}_2\text{O}_4)_2$ and 1 mole of $\text{H}_2\text{C}_2\text{O}_4$ is treated separately with KMnO_4 solution in acidic medium and with $\text{Ba}(\text{OH})_2$ solution having same molarity. Ratio of Volume of KMnO_4 to volume of $\text{Ba}(\text{OH})_2$ consumed in these two titrations is $x:1$, then the value of x is.
59. When 0.5 mole of N_2O_4 is placed in a 4.00L reaction vessel and heated at 405K, 79.3% of the N_2O_4 decomposes to NO_2 . Calculate value of K_p at 405K, for the reaction :
 $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$.
60. Among $[\text{Ni}(\text{CO})_4]$, $[\text{NiCl}_4]^{2-}$, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$, $\text{Na}_3[\text{CoF}_6]$, Na_2O_2 and CsO_2 , the total number of paramagnetic species is

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. Let equation $x^3 - 9x^2 + 24x + \lambda = 0$ has three real and distinct roots a, b, c . If $\lambda \in (-20, -18)$, then the value of $([a] + [b] + [c])$ will be (Where $[.]$ represents greatest integer function.)
- A) 8 B) 6 C) 7 D) 0
62. The equation $ax^3 + bx^2y + cxy^2 + dy^3 = 0$ is a third degree homogenous equation and hence represents three straight lines passing through origin. The condition so that two of these three lines may be perpendicular will be
- A) $(a + d)^2 + ac + bd = 0$ B) $a^2 + ac + bd + d^2 = 0$
C) $a^2 + d^2 + ad + bc = 0$ D) None of these
63. Assertion (A): Let $f(x) = \int_2^x \frac{dt}{\sqrt{1+t^4}}$ and g be the inverse of f , then value of $g'(0)$ is $\sqrt{17}$.
Reason (R): If $g(x)$ is inverse of $f(x)$, then $g(f(x)) = x$.
- A) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
B) Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).
C) Assertion (A) is true and Reason (R) is false.
D) Assertion (A) is false and Reason (R) is true.
64. Let $R = \{(x, y) : x, y \in N \text{ and } x^2 - 4xy + 3y^2 = 0\}$, where N is the set of all natural numbers. Then the relation R is :
- A) reflexive but neither symmetric nor transitive
B) symmetric and transitive
C) reflexive and symmetric
D) reflexive and transitive

65. Let A_1 is a matrix formed by replacing all the elements of a square matrix A of order n by its corresponding cofactors, A_2 is matrix formed by replacing all elements of matrix A_1 by its corresponding cofactors and $A_3, A_4 \dots$ formed so on. If $|A_n| = \lambda$, then $|A|$ equals (where $n > 1$ and A represent determinant value of matrix A)

- A) λ^n B) $\lambda^{(n-1)^n}$ C) $\lambda^{\frac{1}{(n-1)^n}}$ D) $\frac{1}{\lambda^{(n-1)^n}}$

66. Value of $\int_0^{\pi/4} \frac{(\cos \theta - \sin \theta) d\theta}{(\cos \theta + \sin \theta) \sqrt{\sin \theta \cos \theta + \sin^2 \theta \cos^2 \theta}}$ is

- A) $\frac{\pi}{3}$ B) $\frac{\pi}{6}$ C) $\frac{\pi}{4}$ D) $\frac{\pi}{2}$

67. If $\left| z + \frac{1}{z} \right| = 1$ then maximum value of $|z|$ equals

- A) $\frac{3+\sqrt{5}}{2}$ B) $\frac{1+\sqrt{5}}{2}$ C) $\frac{5+\sqrt{3}}{2}$ D) 1

68. The sides of a triangle are roots of the equation $t^3 - 13t^2 + 54t = 72$. If the area of the triangle is λ then 4λ equals

- A) $2\sqrt{455}$ B) $\sqrt{455}$ C) $\sqrt{394}$ D) $2\sqrt{394}$

69.

	LIST-I		LIST-II
(A)	If $a^2 - 4a = 3$, then the value of $\frac{a^3 - a^2 + a - 1}{a^2 - 1}$ is	(I)	3
(B)	The number of values of x satisfying the equation $\sqrt[4]{ x-3 ^{x+1}} - \sqrt[3]{ x-3 ^{x-2}} = 0$ is (if $x \neq 3$)	(II)	2
(C)	The number of value(s) of x satisfying the equation $3^x + 1 - 3^x - 1 = 2 \log_5 6 - x $ is	(III)	4
(D)	The sum of the first $2n$ terms of A.P. 2, 5, 8, ... is equal to the sum of the first n terms of the A.P. 57, 59, 61, ..., then n equal to	(IV)	11

A) (A) – III, (B) – I, (C) – II, (D) – IV

B) (A) – II, (B) – I, (C) – IV, (D) – III

C) (A) – I, (B) – III, (C) – IV, (D) – II

D) (A) – IV, (B) – II, (C) – III, (D) – I

70. If $\vec{a}$ is a unit vector and projection of $\vec{x}$ along $\vec{a}$ is 2 units and $(\vec{a} \times \vec{x}) + \vec{b} = \vec{x}$, then $\vec{x}$ is given by

A) $\frac{1}{2}[\vec{a} - \vec{b} + (\vec{a} \times \vec{b})]$

B) $\frac{1}{2}[2\vec{a} + \vec{b} + (\vec{a} \times \vec{b})]$

C) $[\vec{a} + \vec{a} \times \vec{b}]$

D) $[2\vec{a} - \vec{b} + (\vec{a} \times \vec{b})]$

71. If $\frac{dy}{dx} = f(x) + \int_0^1 f(x)dx$, then the equation of the curve $y = f(x)$ passing through (0,1) is

A) $f(x) = \frac{2e^x - e + 1}{3 - e}$

B) $f(x) = \frac{3e^x - 2e + 1}{2(2 - e)}$

C) $f(x) = \frac{e^x - 2e + 1}{e + 1}$

D) None of these

72. If two tangents from point (3, -2) to the circle $x^2 + y^2 = 1$ cuts x-axis at $A(x_1, 0)$ and $B(x_2, 0)$, then $(x_1 + x_2)$ is equal to

A) 2

B) 10

C) -2

D) 6

73. If $f(x) = \begin{cases} x^3 + 3x^2 + \lambda x + 5 & \text{if } x < 1 \\ 3x - \mu & \text{if } x \geq 1 \end{cases}$ is differentiable for all $x \in R$, then the value of $|\lambda - \mu|$

is

A) 4

B) 6

C) 2

D) 0

74. If $\lambda = \frac{1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots \text{ upto 99 terms}}{1 + 2 + 3 + \dots \text{ upto 100 terms}}$ then λ equals

A) 4983

B) 5050

C) 5151

D) 1

75. If function $f(x)$ satisfies

$$\int f(x) \cos x dx = \frac{1}{2} f^2(x) + c \quad (f(x) \neq 0)$$

Statement-1: The function $f(x)$ satisfies $f(x+\pi) = f(x)$ for all real x .

Statement-2 : $\sin(x+\pi) = -\sin x$ for all real x .

A) Statement-1 is false, statement-2 is true

B) Statement-1 is true, statement-2 is true

C) Statement-1 is false, statement-2 is false

D) Statement-1 is true, statement-2 is false.

76. The line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the curve $xy = c^2, z = 0$ if c equal to

A) ± 1

B) $\pm \frac{1}{3}$

C) $\pm \sqrt{5}$

D) $\pm \sqrt{3}$

77. The area of the region lying between the line $x - y + 2 = 0$, the curve $x = \sqrt{y}$ and y -axis, is (in square units)

A) 9

B) $\frac{9}{2}$

C) $\frac{10}{3}$

D) $\frac{3}{10}$

78. The sum of all values of ' p ' so that the term independent of x in the expansion of $\left(x + \frac{p^2 - 2}{x^4}\right)^{10}$ is 45 is

A) 0

B) 1

C) 2

D) 5

79. Area of quadrilateral formed by joining the foci of the ellipse, $\frac{x^2}{16} + \frac{y^2}{12} = 1$ and $\frac{x^2}{12} + \frac{y^2}{16} = 1$ is (in sq. units)

A) 2

B) 4

C) 6

D) 8

80. Solution of differential equation $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{(1 + \ln x + \ln y)}$ is (where C is integration constant)

A) $2(1 + \ln(xy)) = x^2 + C$

B) $xy \ln(xy) = x^2 + C$

C) $(1 + \ln(xy)) = x^2 + C$

D) $2xy \ln(xy) = x^2 + C$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. If the number of integral values of x , satisfying the inequality $\log_{\frac{1}{x}} \left(\frac{2(x-2)}{(x+1)(x-5)} \right) \geq 1$ is λ , then λ equals
82. Let $f(k) = \frac{k}{2009}$ and $g(k) = \frac{(f(k))^4}{(1-f(k))^4 + (f(k))^4}$, then the sum $\sum_{k=0}^{2009} g(k)$ is equal to
83. If $\sum_{r=1}^{20} \tan^{-1} \left(\frac{1}{2r^2} \right) = \tan^{-1} \frac{m}{n}$; where m and n are co-prime numbers then $[\log_2(m+n)]$ equals (where $[.]$ is greatest integer function)
84. Let $\sin a + 2\sin b + 3\sin c = 0$ and $\cos a + 2\cos b + 3\cos c = 0$. If the value of $\cos 3a + 8\cos 3b + 27\cos 3c = \lambda \cos(a+b+c)$ then λ equals
85. If $\cot 7\frac{1}{2}^\circ = \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3} + \sqrt{n_4}$ (where n_1, n_2, n_3, n_4 are natural numbers) then the value of $\sum_{i=1}^4 n_i$
86. Smallest possible natural number ' n ' for which the equation $x^2 - nx + 2014 = 0$ has integer roots, is
87. Let A be the set of 5 digit number formed by digits 1, 2, 3, 4, 5 using each digit exactly once, such that exactly two odd positions are occupied by odd digit. If the sum of digit in the rightmost position of the numbers in A is λ then the remainder when λ is divided by 7 is
88. If 1 lies between the roots of equation $x^2 - px + 3 = 0$ and $t \in R$ then the value of $\left[\left(\frac{2|t|}{|t|^2 + 4} \right)^p \right]$ is (where $[.]$ denotes greatest integer function)
89. If $f(x) = \frac{(2x+5)}{(x-3)}$ and $f(g(x)) = \frac{1}{(x+1)}$, then $\left| \frac{9g(4)}{14} \right|$ is
90. A point P moves such that slope of one tangent drawn from it to parabola $y^2 = 16x$ is four times the slope of other tangent, then the length of latus rectum of the conic represented by locus of P , is